CAUSE-BENCH: EVALUATING SELECTIVE SOUND GENERATION THROUGH VISUAL ACTION DEACTIVATION

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ABSTRACT

Video-to-audio (V2A) generation aims to synthesize plausible and temporally aligned audio from visual content. However, current V2A generation and evaluation still face three limitations. First, existing models can generate sounds for visible sources even when the corresponding sounding actions are inactive. Second, dedicated benchmarks for evaluating this failure are lacking. Third, conventional quality and alignment metrics do not explicitly measure such event selectivity. To tackle these limitations, we make three contributions. First, we introduce CauSE-Bench, containing 102 original–edited video pairs across nine sound-event classes. Each edit deactivates one selected action while preserving its visible source and other sounding actions. Second, we propose CauSE-Acc, a class-balanced paired metric for evaluating selective sound generation. Third, we present AudioEval, a common toolbox for complementary audio and audiovisual evaluation. Across six V2A models, the highest automatic CauSE-Acc is 29.43%, showing that selective event generation remains challenging under this protocol. These results motivate evaluating event selectivity alongside audio quality and synchronization. Our dataset, evaluation code, and demos are available at `https://cause-bench.github.io/`.

***Index Terms*—** video-to-audio generation, audiovisual benchmarking, sound event detection, audio evaluation

1. INTRODUCTION

Video-to-audio (V2A) generation should produce sounds that match the actions shown in a video. For example, if a dog in the video keeps its mouth closed, the generated audio should not contain barking. Similarly, if a hammer never hits anything, the generated audio should not contain impact sounds. Generating such sounds merely because their sources are visible is a form of audio hallucination. We investigate whether V2A models generate a sound when the corresponding action occurs, and suppress that sound when the action is absent. Distinguishing these two cases is challenging because the visual source is present in both, and only its action changes. **A reliable V2A system should therefore respond to action-level visual evidence rather than source identity alone.**

Recent V2A systems, including Diff-Foley [1], MMAudio [2], HunyuanVideo-Foley [3], and FoleyCrafter [4], have improved audio quality and audiovisual alignment. Evaluations commonly use VGGSound [5] and Movie Gen Audio Bench [6], with metrics for audio quality (e.g., FAD [7]), semantic alignment (e.g., ImageBind similarity [8]), and synchronization (e.g., Synchformer-based scores [9]). HALCON [10] evaluates insertion hallucinations, including generated speech or music without corresponding visible sources. FlatSounds [11] uses controlled video pairs to test physical properties of generated sound. These studies motivate testing specific failures beyond overall audio quality.

However, evaluating whether a visible but inactive source incorrectly triggers sound requires a more specific comparison. Natural test clips do not provide matched active and inactive versions of the same event. Controlled pairs isolate this behavior by changing one sound-producing action while keeping the source, surrounding scene, and other sounding events unchanged. Standard quality and alignment scores also do not jointly require target generation, target suppression, and preservation of other sounds. Finally, comparisons depend on choices such as feature extractors, audio preprocessing, and clip selection; a common implementation and explicit settings improve reproducibility.

To address these limitations, we make three contributions. First, we introduce **CauSE-Bench**, a benchmark and semi-automatic construction pipeline that deactivates one visible sound-producing action while preserving its source and other sounding actions. Second, we propose **CauSE-Acc**, a paired metric that jointly evaluates target generation, target suppression, and shared-event retention. Third, we present **AudioEval**, an open-source toolbox for audio and audiovisual evaluation, and evaluate six V2A models using the same benchmark and scoring protocol.

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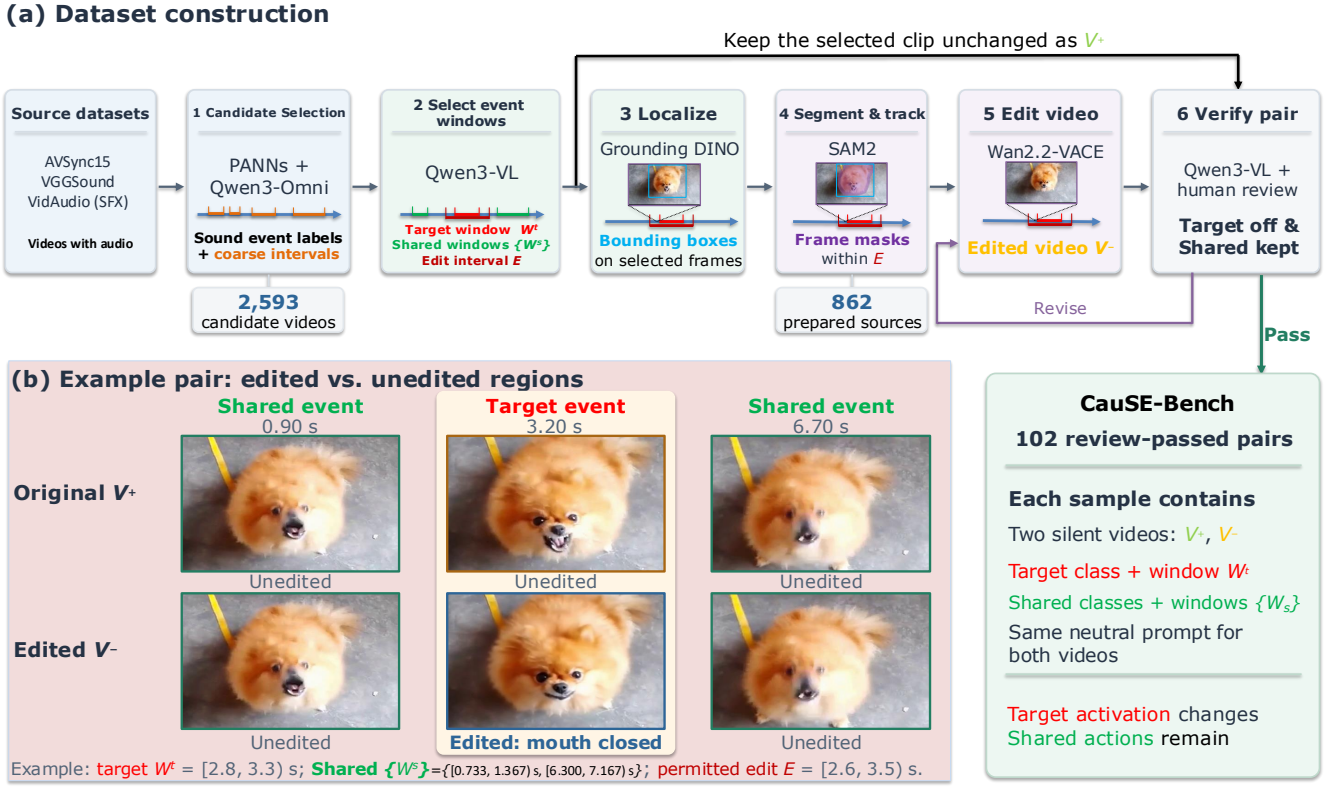


Fig. 1. Overview of CauSE-Bench. (a) Six-stage construction pipeline: candidate events are refined into the target window W^t and shared windows $\{W^s\}$; the target source is then localized and segmented, V^+ is edited within E around W^t to produce V^-, and the resulting pair is verified, while shared windows remain unedited. (b) Example factual–counterfactual pair.

2. BENCHMARK CONSTRUCTION

Let $V^+ \in \mathbb{R}^{T\times 3\times H\times W}$ denote the silent original video and $Y \in \mathbb{R}^L$ its recorded audio, where T, H, and W denote the frame count, height, and width, respectively, and L is the number of audio samples. The pipeline in Fig. 1 produces $V^- \in \mathbb{R}^{T\times 3\times H\times W}$ with the same duration as V^+. V^+ and V^- differ only within an editing interval E, where one sound-producing action is deactivated while its visible source and shared actions are preserved. The pair (V^+, V^-) is used for the evaluation in Section 3.

2.1. Candidate selection (Fig. 1(a-1))

We collect clips from AVSync15 [12], VGGSound [5], and the SFX subset of VidAudio-Bench [13]. PANNs [14] proposes candidate sound events with sound classes and coarse temporal boundaries, while Qwen3-Omni [15] verifies that the events are audible and temporally separable and that their sound sources are visible in sampled frames.

For each retained clip, this stage outputs candidate windows $\mathcal{W} = \{W_1, \ldots, W_N\}$, as shown in orange color in Fig. 1(a-1). Each $W_i = [s_i, e_i)$ has an associated sound class c_i, with $s_i < e_i$ measured in seconds on the video timeline.

2.2. Select event windows (Fig. 1(a-2))

Given video V^+, candidate windows $\mathcal{W}$ with sound classes, Qwen3-VL [16] verifies the corresponding visible sound-producing actions and refines the window boundaries. From the updated $\mathcal{W}$, we select one target window $W^t \in \mathcal{W}$ for action deactivation and $K \geq 1$ shared windows $\{W^s_j\}_{j=1}^K \subset \mathcal{W} \setminus \{W^t\}$ whose actions must remain unchanged. Qwen3-VL also outputs short noun queries for the sound source and coupled objects, such as "dog", for spatial localization in 2.3. We define an editing interval $E \supseteq W^t$ such that $E \cap W^s_j = \varnothing$ for all j. While W^t specifies the target event for audio evaluation, E may extend around W^t so that the sound-producing action can transition naturally into and out of the inactive state. In Fig. 1(b), $W^t = [2.8, 3.3)$ s and $E = [2.6, 3.5)$ s.

2.3. Localize the source in video frames (Fig. 1(a-3))

Grounding DINO [17] takes selected video frames and the object names from 2.2 as input and outputs bounding boxes around these objects. For example, the query "dog" produces a box around the dog in each selected frame. These boxes guide SAM2 to segment the objects and track their masks across frames in the next stage.

2.4. Segment and track image masks (Fig. 1(a-4))

SAM2 [18] takes V^+ and the bounding boxes from 2.3 as input to segment the objects and propagate their masks across frames. We adjust these masks to cover the sound-producing mechanism and necessary coupled motion, then restrict editing to E. The output is a sequence of spatial editing masks; regions outside them are preserved.

2.5. Deactivate the target action (Fig. 1(a-5))

Wan2.2-VACE-Fun-A14B[1], based on VACE [19], takes V^+, the editing masks, and an inactive-state instruction as input to generate edited content. For the example in Fig. 1(b), the dog barks during W^t; the instruction asks the model to deactivate this action by keeping its mouth closed. We use 40 sampling steps and may additionally provide a verified resting-state reference. The generated content replaces the corresponding regions in V^+ only within the editing masks during E, while all content outside these masks is kept unchanged. This produces V^- with the same frame layout and duration as V^+.

2.6. Verify and retain pairs (Fig. 1(a-6))

We use Qwen3-VL to compare V^+ and V^- and check whether the target action is deactivated, shared actions remain, the source identity is preserved, and motion remains natural. Human annotators then review these checks, inspect editing artifacts, and confirm the temporal boundaries of the target and shared windows. The original audio Y serves as a factual reference during this review. Failed or ambiguous edits are revised or discarded. Each accepted item contains the video pair (V^+, V^-), the verified target window W^t, and shared windows $\{W_j^s\}_{j=1}^K$.

The construction audit records 2,593 candidate videos, of which 862 have prepared editing inputs. The final evaluation set contains 102 source-distinct video pairs across nine classes: dog barking (35), lion roaring (16), machine-gun shooting (11), baby vocalizations (9), cap-gun shooting (8), hammering (8), frog croaking (7), chicken crowing (5), and knife sharpening (3).

3. EVALUATION AND METRICS

3.1. Paired evaluation and CauSE-Acc

We apply the evaluated video-to-audio (V2A) generation models to V^+ and V^- under identical inference settings, obtaining the corresponding audio outputs $\hat{Y}^+$ and $\hat{Y}^-$. We crop each waveform to its input video duration, convert it to mono, and resample it to 32 kHz. We then use PANNs Cnn14 [14] (checkpoint mAP: 0.431) for sound-event scoring.

For each annotated window W_i with sound class c_i, we define $S(\hat{Y}, W_i) \in [0,1]$ as the probability predicted by PANNs that c_i is present in the audio segment associated with W_i. The corresponding binary indicator is

$$D(\hat{Y}, W_i) = \mathbf{1}\Big[S(\hat{Y}, W_i) \geq \tau\Big], \quad \tau = 0.1, \tag{1}$$

where $\mathbf{1}[\cdot]$ is the indicator function. Thus, $D(\hat{Y}, W_i) = 1$ indicates that class c_i is detected in the associated audio segment, whereas $D(\hat{Y}, W_i) = 0$ indicates that it is not detected.

For each generated pair $(\hat{Y}^+, \hat{Y}^-)$, we evaluate three binary conditions: **Hit** detects the target event in $\hat{Y}^+$ within W^t; **Reject** requires its absence from $\hat{Y}^-$ within W^t; and **Keep** detects every shared event in both outputs within its corresponding window:

$$\begin{aligned} &\mathrm{Hit} = D(\hat{Y}^+, W^t), \qquad \mathrm{Reject} = 1 - D(\hat{Y}^-, W^t), \\ &\mathrm{Keep} = \prod_{j=1}^{K} D(\hat{Y}^+, W_j^s) D(\hat{Y}^-, W_j^s). \end{aligned} \tag{2}$$

For each generated pair, $\mathrm{Hit}, \mathrm{Reject}, \mathrm{Keep} \in \{0,1\}$, with 1 indicating the desired behavior. Their product, $\mathrm{PairSuccess} = \mathrm{Hit} \times \mathrm{Reject} \times \mathrm{Keep}$, equals 1 only when all three conditions are satisfied. **CauSE-Acc** aggregates $\mathrm{PairSuccess}$ in a class-balanced manner: we first average over pairs within each target class and then equally across classes. Hit, Reject, and Keep use the same aggregation. $\mathrm{PairSuccess}$ is computed per pair before aggregation. Overall, $\mathrm{CauSE\text{-}Acc}$ summarizes selective event-generation performance, with higher values indicating better performance.

3.2. AudioEval toolbox and complementary metrics

Reproducible comparison requires explicit choices of feature extractor, preprocessing, reference data, and temporal sampling. We also release `audio_eval`[2], an open-source toolbox for evaluating audio generation and reconstruction. It provides commonly used metrics through a unified command-line interface and Python API, enabling multiple metrics to be computed with one command. Beyond V2A, `audio_eval` supports text-to-audio, text-to-music, speech synthesis, audio reconstruction, and audio super-resolution.

For V2A, we report six complementary metrics. FAD-VGGish [7] and FD-PANNs measure Fréchet distances using VGGish and PANNs embeddings, respectively. KL and Inception Score (IS) use normalized PANNs class scores [14]; KL is computed in the reference-to-generated direction, $D_{\mathrm{KL}}(p_{\mathrm{GT}} \| p_{\mathrm{gen}})$. ImageBind similarity (IB) [8] measures audiovisual semantic correspondence. DeSync uses Synchformer [9] to estimate the absolute temporal offset in seconds, averaged over the first and last 4.8-second windows. These metrics assess distributional similarity, semantics, or synchronization; but not the three paired checks above.

[1] https://huggingface.co/alibaba-pai/Wan2.2-VACE-Fun-A14B

[2] https://github.com/shwj114514/audio_eval

Table 1. Results on CauSE-Bench. Supplementary metrics are computed on factual outputs only. Best model results are shown in bold; GT is excluded.

	CauSE metrics (ours, %)				Audio metrics				AV metrics	
Model	**Hit ↑**	**Reject ↑**	**Keep ↑**	**CauSE-Acc ↑**	FAD ↓	FD ↓	KL ↓	IS ↑	IB(%) ↑	DeSync(s) ↓
Original GT audio	98.99	–	93.90	–	0	0	0	4.835	41.88	0.197
Diff-Foley	34.33	**82.60**	24.22	9.52	7.754	56.761	1.772	4.299	28.45	0.955
HunyuanVideo-Foley	42.43	81.02	24.35	10.26	6.977	52.645	1.716	4.094	29.11	**0.272**
FoleyCrafter	52.58	53.31	39.29	8.87	6.481	50.218	1.153	4.117	36.03	1.124
MMAudio (medium_44k)	**74.17**	63.21	**56.87**	**29.43**	**2.307**	**24.917**	**0.554**	4.503	**39.51**	0.274
AudioX-Turbo	55.69	63.70	37.82	8.13	5.535	34.826	1.116	**4.847**	33.14	1.057
UniFlow-Audio (v1.1 Base)	55.31	78.81	32.38	14.59	10.886	45.533	1.380	4.556	31.56	0.348

4. EXPERIMENTS

4.1. Experimental Setup

We evaluate six publicly available models: Diff-Foley [1], HunyuanVideo-Foley [3] (XXL), FoleyCrafter [4], MMAudio [2] (medium_44k), AudioX-Turbo [20], and UniFlow-Audio [21] (v1.1 Base). Each generates 204 soundtracks for the 102 pairs using silent video inputs, without generated captions. We follow the official inference implementations and publicly released configurations for all models. Diff-Foley uses 25 sampling steps with classifier-free guidance (CFG) 4.5 and classifier guidance 50; HunyuanVideo-Foley uses 50 steps with CFG 4.5; FoleyCrafter uses 25 steps with CFG 7.5; MMAudio uses 25 steps with CFG 4.5; AudioX-Turbo uses four steps without CFG and UniFlow-Audio uses 25 steps with CFG 5. All generated soundtracks are cropped to the corresponding video duration.

As a detector sanity check, we also evaluate the original recorded audio on the annotated target and shared windows, obtaining detection rates of 98.99% and 93.90%, respectively.

4.2. Main comparison and metric validity

Table 1 shows that MMAudio leads in CauSE-Acc (29.43%), Hit (74.17%), and Keep (56.87%). UniFlow-Audio ranks second in CauSE-Acc (14.59%). Diff-Foley obtains the highest Reject (82.60%) but low Hit (34.33%) and Keep (24.22%), yielding only 9.52% CauSE-Acc. High rejection alone therefore does not establish successful selective generation.

MMAudio also achieves the best FAD, FD, KL, and IB on original-video outputs. However, HunyuanVideo-Foley and MMAudio have similar DeSync scores (0.272 and 0.274 s) despite different CauSE-Acc scores (10.26% and 29.43%). Similar synchronization scores therefore do not imply comparable paired event correctness.

Among the complementary metrics, MMAudio achieves the lowest FAD, FD, and KL and the highest ImageBind similarity. However, these global semantic and synchronization metrics do not necessarily reflect action-level event correctness. As shown in Fig. 2, the edited-video output of MMAudio achieves higher IB and lower DeSync than the original-video output, yet still generates crowing after the target action is visually deactivated. CauSE-Acc captures this failure through $\mathrm{Reject} = 0$ and $\mathrm{PairSuccess} = 0$, showing that strong global audiovisual alignment does not guarantee sensitivity to whether a sound-producing action is actually active.

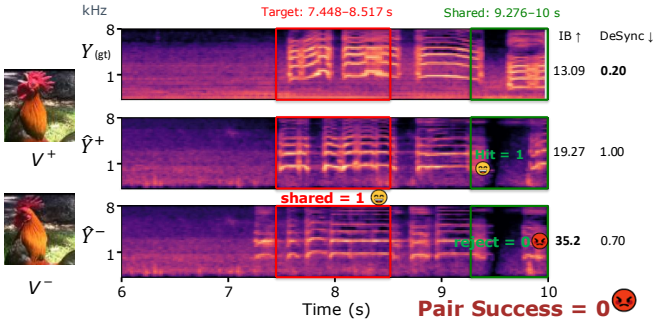


Fig. 2. MMAudio case study. Although $\hat{Y}^-$ achieves better IB and DeSync, it still generates crowing in the deactivated target window (red), yielding Reject$= 0$ and PairSuccess$= 0$. The shared event (green) is preserved.

5. CONCLUSION

We introduced CauSE-Bench for evaluating whether V2A systems respond selectively to the activation state of visible sound-producing actions. CauSE-Acc jointly measures target generation, target suppression, and shared-event preservation. Our results show that strong global semantic alignment or synchronization does not necessarily imply correct responses to local action states. CauSE-Bench therefore complements conventional V2A metrics with an explicit test of event selectivity.

6. ACKNOWLEDGMENT

AI tools were used for language polishing and did not contribute to the technical content of this work.

7. COMPLIANCE WITH ETHICAL STANDARDS

This study did not require ethical approval under our institutional guidelines.

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